Systems of Linear Equations — Full Solutions

Digital SAT Style | Every Step Fully Explained

Q1. Correct Answer: B) 3

STEP-BY-STEP SOLUTION:

Add the two equations (the ±2y cancels):

$(3x+2y)+(4x-2y)=14+7$

$7x = 21 \rightarrow x = 3$

Verify: $3(3)+2y=14 \rightarrow y=2.5$. Check eq2: $4(3)-2(2.5)=12-5=7$ ✓

WHY EACH OPTION IS WRONG/RIGHT:

A) 2 — Dividing 21 by 7 incorrectly.

B) 3 — CORRECT.

C) 4 — Adding 14+7=21 but dividing by wrong coefficient.

D) 5 — No valid algebraic path leads here.

🪤 **TRAP ALERT:** Trying substitution when elimination is instant. The ±2y terms cancel directly — always look for this.

💡 **KEY INSIGHT:** When y-coefficients are equal and opposite, ADD equations. Same sign → SUBTRACT.

⚡ **FAST METHOD:** 💻 DESMOS: Type both equations. Click intersection. x-coordinate = 3.

Q2. Correct Answer: A)

STEP-BY-STEP SOLUTION:

Define: s = standard, p = premium.

Count equation: $s + p = 35$

Revenue equation: $12s + 20p = 580$

This matches option A.

Verify: $s=15, p=20 \rightarrow 15+20=35$ ✓, $12(15)+20(20)=180+400=580$ ✓

WHY EACH OPTION IS WRONG/RIGHT:

A) CORRECT.

B) Swaps the two equations completely.

C) Reverses which price belongs to which ticket.

D) Same error as B — equations are swapped.

🪤 **TRAP ALERT:** Assigning the wrong price to the wrong ticket type. Re-read which costs $12 and which costs $20.

💡 **KEY INSIGHT:** Write the simpler count equation first, then the money equation. Match coefficients to prices carefully.

⚡ **FAST METHOD:** No calculator needed. Read carefully and match each ticket's price to its variable.

Q3. Correct Answer: D) 4

STEP-BY-STEP SOLUTION:

Add the two equations (y and −y cancel):

$(5x-y)+(x+y) = 11+7$

$6x = 18 \rightarrow x = 3$

Substitute into $x+y=7$: $3+y=7 \rightarrow y=4$

WHY EACH OPTION IS WRONG/RIGHT:

A) 1 — Confused x=3 and subtracted from 4 instead of 7.
B) 2 — Substitution arithmetic error.
C) 3 — Wrote x=3 as the answer but question asks for y.
D) 4 — CORRECT.

🪤 **TRAP ALERT:** The question asks for y, but elimination gives x first. Students stop at x=3 and pick C. Always re-read what is being asked.

💡 **KEY INSIGHT:** Circle what the question asks for BEFORE you start solving. Here: y, not x.

⚡ **FAST METHOD:** 💻 DESMOS: Graph both lines, click intersection → y-coordinate = 4.

Q4. Correct Answer: B) 3

STEP-BY-STEP SOLUTION:

Line A through (0,5) and (5,0):

Slope = $(0-5)/(5-0) = -1$

Equation: $y = -x + 5$

Set equal to Line B ($y=2x-4$):

$-x + 5 = 2x - 4$

$9 = 3x \rightarrow x = 3$

Verify: $y = -3+5 = 2$ and $y = 2(3)-4 = 2$ ✓

WHY EACH OPTION IS WRONG/RIGHT:

A) 2 — Computed slope incorrectly as −2.
B) 3 — CORRECT.
C) 4 — Used x-intercept of Line A instead.
D) 5 — Picked the x-intercept of Line A.

🪤 **TRAP ALERT:** Not finding Line A's equation from the two points before setting equal to Line B.

💡 **KEY INSIGHT:** Two-point line: find slope first → use $y = mx + b$ with one point → set equal to the other line.

⚡ **FAST METHOD:** 💻 DESMOS: Type $y=-x+5$ and $y=2x-4$. Click intersection → x=3.

Q5. Correct Answer: B) 4

STEP-BY-STEP SOLUTION:

Both equations have 4x. Subtract the second from the first:

$(4x+3y)-(4x-y) = 24-8$

$4y = 16 \rightarrow y = 4$

Verify: $4x-4=8 \rightarrow 4x=12 \rightarrow x=3$. Check: $4(3)+3(4)=24$ ✓

WHY EACH OPTION IS WRONG/RIGHT:

A) 2 — Divided 8 by 4 (from second equation alone).
B) 4 — CORRECT.
C) 5 — Arithmetic error in subtraction.
D) 6 — Sign error: treated −y as +y when subtracting.

🪤 **TRAP ALERT:** When subtracting equations, the −y in the second becomes +y. Write it out: $3y-(-y) = 3y+y = 4y$.

💡 **KEY INSIGHT:** Same coefficient with same sign → SUBTRACT. Watch every sign when subtracting.

⚡ **FAST METHOD:** Subtraction eliminates x instantly: $4y=16 \rightarrow y=4$.

Q6. Correct Answer: C) 3.5

STEP-BY-STEP SOLUTION:

Layla: $L = 15w + 45$
Marcus: $M = 25w + 10$
Set equal: $15w + 45 = 25w + 10$
$35 = 10w \rightarrow w = 3.5$
Verify: L=15(3.5)+45=$97.50. M=25(3.5)+10=$97.50 ✓

WHY EACH OPTION IS WRONG/RIGHT:

A) 2.5 — Forgot starting balances.
B) 3 — At w=3: L=90, M=85. Not equal.
C) 3.5 — CORRECT.
D) 4 — At w=4: L=105, M=110. Marcus has more.

🪤 **TRAP ALERT:** Forgetting the starting balances ($45 and $10). Without them you get $15w=25w$ which has no solution.

💡 **KEY INSIGHT:** Rate × time + starting amount = total. Always include the initial value.

⚡ **FAST METHOD:** 💻 DESMOS: Graph $y=15x+45$ and $y=25x+10$. Intersection at $x=3.5$.

Q7. Correct Answer: B) (2, 4)

STEP-BY-STEP SOLUTION:

Both equal y. Set equal:
$3x - 2 = -x + 6$
$4x = 8 \rightarrow x = 2$
$y = 3(2)-2 = 4$
Solution: (2, 4)

WHY EACH OPTION IS WRONG/RIGHT:

A) (1,1) — $y=3(1)-2=1$, but $y=-1+6=5$. Not equal.
B) (2,4) — CORRECT: both give $y=4$.
C) (3,3) — $y=3(3)-2=7\neq3$.
D) (4,2) — Flipped x and y from the correct answer.

🪤 **TRAP ALERT:** Finding $x=2$ and $y=4$, then accidentally writing (4,2) by swapping coordinates.

💡 **KEY INSIGHT:** Verify BOTH equations with your answer. Takes 10 seconds and catches errors.

⚡ **FAST METHOD:** 💻 DESMOS: Enter both equations. Click intersection.

Q8. Correct Answer: A) 5

STEP-BY-STEP SOLUTION:

Add equations: $(2x+y)+(x-y) = 9+3$

$3x = 12 \rightarrow x = 4$

Substitute: $4-y=3 \rightarrow y=1$

$x + y = 4 + 1 = 5$

WHY EACH OPTION IS WRONG/RIGHT:

A) 5 — CORRECT.

B) 6 — Substitution error getting $y=2$.

C) 7 — Confused $2x+y=9$ with $x+y=7$.

D) 8 — Subtracted y from 9 instead of solving properly.

🪤 **TRAP ALERT:** Reading $2x+y=9$ and thinking $x+y$ must equal something close to 9. The 2x makes a big difference.

💡 **KEY INSIGHT:** The question asks for $x+y$, not x or y alone. Solve for both, then compute the expression.

⚡ **FAST METHOD:** Add → $x=4$. Sub back → $y=1$. Sum=5.

Q9. Correct Answer: B) 3

STEP-BY-STEP SOLUTION:

$3.50 + 1.75m = 5.00 + 1.25m$

$0.50m = 1.50$

$m = 3$

Verify: Taxi=3.50+5.25=$8.75. Rideshare=5.00+3.75=$8.75 ✓

WHY EACH OPTION IS WRONG/RIGHT:

A) 2 — Taxi=$7.00, Rideshare=$7.50. Not equal.

B) 3 — CORRECT.

C) 4 — Taxi=$10.50, Rideshare=$10.00. Not equal.

D) 5 — Taxi overtakes rideshare at this point.

🪤 **TRAP ALERT:** Dividing flat-fee difference by total rate instead of rate difference: $(5-3.5)/(1.75+1.25)=0.5 \neq$ correct.

💡 **KEY INSIGHT:** Break-even: difference in fixed costs ÷ difference in per-unit rates = $1.50/0.50 = 3$.

⚡ **FAST METHOD:** 💻 DESMOS: $y=3.5+1.75x$ and $y=5+1.25x$. Intersection at $x=3$.

Q10. Correct Answer: B) 2

STEP-BY-STEP SOLUTION:

Subtract second from first:

(6x+2y)−(3x+2y) = 20−14

3x = 6 → x = 2

WHY EACH OPTION IS WRONG/RIGHT:

A) 1 — Divided 6 by 6 instead of 3.

B) 2 — CORRECT.

C) 3 — Divided 6 by 2 instead of 3.

D) 4 — Added equations instead of subtracting.

🪤 **TRAP ALERT:** Adding when you should subtract. Same-sign same-coefficient → SUBTRACT.

💡 **KEY INSIGHT:** Matching coefficients with same sign → subtract to eliminate.

⚡ **FAST METHOD:** 3x=6→x=2. One step.

Q11. Correct Answer: B) 4

STEP-BY-STEP SOLUTION:

For infinitely many solutions, eq2 must be a multiple of eq1.

Ratio: 6/3 = 2, so multiply eq1 by 2: 6x+2ky=24.

This must equal 6x+8y=24.

So 2k = 8 → k = 4.

Cross-check constant: 12×2=24 ✓

WHY EACH OPTION IS WRONG/RIGHT:

A) 2 — Would give different slopes → unique solution.

B) 4 — CORRECT.

C) 6 — Confused k with the multiplier.

D) 8 — Used the coefficient of y from eq2, not solving for k.

🪤 **TRAP ALERT:** Picking k=8 because you see 8 in equation 2. You need to SOLVE 2k=8, not copy 8.

💡 **KEY INSIGHT:** Infinitely many: $a_1/a_2 = b_1/b_2 = c_1/c_2$. Set up proportion: 3/6 = k/8 → k=4.

⚡ **FAST METHOD:** k/8 = 3/6 = 1/2 → k=4. Done in seconds.

Q12. Correct Answer: C) 8

STEP-BY-STEP SOLUTION:

Let m=muffins, c=croissants.

Flour: 0.5m + 1.5c = 15 → multiply by 2: m + 3c = 30 ...(1)

Eggs: m + 2c = 22 ...(2)

Subtract (2) from (1): c = 8

m = 22 − 2(8) = 6

Verify flour: 0.5(6)+1.5(8) = 3+12 = 15 ✓

Verify eggs: 6+2(8) = 6+16 = 22 ✓

WHY EACH OPTION IS WRONG/RIGHT:

A) 4 — Solving a misread equation.

B) 6 — Found m (muffins) instead of c (croissants).

C) 8 — CORRECT.

D) 10 — Egg constraint violated: 6+20=26≠22.

🪤 **TRAP ALERT:** Solving for m and reporting it as the answer when the question asks for croissants (c). Re-read the question.

💡 **KEY INSIGHT:** Clear decimals first (multiply flour equation by 2), then subtract cleanly.

⚡ **FAST METHOD:** After clearing decimals: m+3c=30 and m+2c=22. Subtract → c=8.

Q13. Correct Answer: A) −10

STEP-BY-STEP SOLUTION:

For no solution: same slopes, different intercepts.

Coefficient ratios must be equal: 2/4 = (−5)/p

1/2 = −5/p → p = −10

Verify constants are NOT proportional: 3/9 = 1/3 ≠ 1/2 ✓

Since coefficient ratio (1/2) ≠ constant ratio (1/3), the system has NO solution.

WHY EACH OPTION IS WRONG/RIGHT:

A) −10 — CORRECT.

B) −5 — Ratio: −5/−5=1 ≠ 1/2. Different slopes → one solution.

C) 5 — Ratio: −5/5=−1 ≠ 1/2. One solution.

D) 10 — Ratio: −5/10=−1/2 ≠ 1/2. One solution.

🪤 **TRAP ALERT:** Confusing no solution with infinitely many. No solution: $a_1/a_2 = b_1/b_2 \neq c_1/c_2$. Infinitely many: ALL ratios equal.

💡 **KEY INSIGHT:** No solution = parallel lines = same slope, different y-intercept. Set coefficient ratios equal, then verify constant ratio differs.

⚡ **FAST METHOD:** 2/4 = −5/p → p=−10. Check: 3/9 ≠ 1/2 ✓. No solution confirmed.

Q14. Correct Answer: 648

STEP-BY-STEP SOLUTION:

a + b = 54 ...(1)

a − b = 18 ...(2)

Add: 2a = 72 → a = 36

From (1): b = 54 − 36 = 18

Product = 36 × 18 = 648

WHY EACH OPTION IS WRONG/RIGHT:

972 — Multiplied 54×18 (the given sum and difference).

36 — Stopped after finding only one number.

54 — Used the sum itself.

648 — CORRECT.

🪤 **TRAP ALERT:** Stopping after finding a=36 and b=18 without computing the PRODUCT. DSAT grid-ins always have one more step.

💡 **KEY INSIGHT:** Multi-step grid-ins: the question never asks for the first thing you solve. Always complete the final calculation.

⚡ **FAST METHOD:** a=(54+18)/2=36. b=(54−18)/2=18. Product=648.

Q15. Correct Answer: C) 49

STEP-BY-STEP SOLUTION:

For infinitely many: eq2 = k × eq1.
Find multiplier using y-coefficients: 12/4 = 3.
So: $3 \times (ax + 4y = 16) = 3ax + 12y = 48$.
This must equal $3x + 12y = c$.
$3a = 3 \rightarrow a = 1$
$c = 48$
$a + c = 1 + 48 = 49$

WHY EACH OPTION IS WRONG/RIGHT:

A) 4 — Only found a=1, added wrong c value.
B) 16 — Used the constant from eq1.
C) 49 — CORRECT.
D) 64 — Set a=16 instead of solving 3a=3.

🪤 **TRAP ALERT:** Finding multiplier=3 correctly but applying it to get a wrong value for c.

💡 **KEY INSIGHT:** Multiplier = 12/4 = 3. Apply to ALL terms: a=3/3=1, c=16×3=48.

⚡ **FAST METHOD:** Multiplier=3. a=1, c=48. Sum=49.

Q16. Correct Answer: A) 60

STEP-BY-STEP SOLUTION:

$b + p = 200$ ...(1)
$8b + 15p = 2{,}020$ ...(2)
From (1): $b = 200-p$. Substitute into (2):
$8(200-p) + 15p = 2{,}020$
$1{,}600 - 8p + 15p = 2{,}020$
$7p = 420 \rightarrow p = 60$

WHY EACH OPTION IS WRONG/RIGHT:

A) 60 — CORRECT.
B) 80 — Check: 8(120)+15(80) = 960+1200 = \$2,160 ≠ \$2,020.
C) 100 — Check: 8(100)+15(100) = \$2,300.
D) 140 — This is the number of BASIC memberships.

🪤 **TRAP ALERT:** Solving for b (basic) when the question asks for p (premium). b=140, p=60.

💡 **KEY INSIGHT:** After solving, always check which variable the question asks for.

⚡ **FAST METHOD:** 💻 DESMOS: Graph b+p=200 and 8b+15p=2020. Find p at intersection.

Q17. Correct Answer: A) 10

STEP-BY-STEP SOLUTION:

Add equations (±2b cancels):
$(3a-2b)+(a+2b) = 11+9$
$4a = 20 \rightarrow a = 5$
$2a = 10$

WHY EACH OPTION IS WRONG/RIGHT:

A) 10 — CORRECT.
B) 12 — Divided 20 by 4 as ≈5.something, then doubled.
C) 14 — Added 5+9=14 instead of computing 2(5).
D) 16 — Forgot to divide 20 by 4 first.

🪤 **TRAP ALERT:** Finding a=5 and answering 5 instead of 2a=10. The question asks for 2a, not a.

💡 **KEY INSIGHT:** DSAT often asks for expressions like 2a or x+y, not individual values. Read carefully.

⚡ **FAST METHOD:** Add equations → 4a=20 → a=5 → 2a=10. No need to find b.

Q18. Correct Answer: C) 8

STEP-BY-STEP SOLUTION:

Height A after t hours: H_A = 24 − 2t
Height B after t hours: H_B = 16 − t
Set equal: 24 − 2t = 16 − t
8 = t → t = 8 hours
Verify: A=24−16=8cm. B=16−8=8cm ✓

WHY EACH OPTION IS WRONG/RIGHT:

A) 4 — A=16, B=12. Not equal.
B) 6 — A=12, B=10. Not equal.
C) 8 — CORRECT. Both = 8cm.
D) 10 — A=4, B=6. Not equal.

🪤 **TRAP ALERT:** Writing H = start + rate×t instead of start − rate×t. Candles DECREASE in height.

💡 **KEY INSIGHT:** Decreasing quantities: H = initial − rate × t. The minus sign is critical.

⚡ **FAST METHOD:** 💻 DESMOS: y=24−2x and y=16−x. Intersection at x=8.

Q19. Correct Answer: A) 1/5

STEP-BY-STEP SOLUTION:

For infinitely many solutions: all ratios equal.
Constant ratio: 12/30 = 2/5
From m/5 = 2/5 → m = 2
From 4/n = 2/5 → 2n = 20 → n = 10
m/n = 2/10 = 1/5

WHY EACH OPTION IS WRONG/RIGHT:

A) 1/5 — CORRECT.
B) 2/5 — This is the common ratio, not m/n.
C) 5/2 — Inverted the ratio.
D) 5 — Confused n/m with m/n.

🪤 **TRAP ALERT:** Computing the common ratio (2/5) and picking that as the answer instead of computing m/n.

💡 **KEY INSIGHT:** Use the constant ratio to find the multiplier, then find each variable separately, THEN compute m/n.

⚡ **FAST METHOD:** Ratio=12/30=2/5. m=2, n=10. m/n=1/5.

Q20. Correct Answer: 4

STEP-BY-STEP SOLUTION:

C_A = 50 + 30m
C_B = 10 + 40m
Set equal: 50+30m = 10+40m
40 = 10m → m = 4
Verify: A=50+120=$170. B=10+160=$170 ✓

WHY EACH OPTION IS WRONG/RIGHT:

6 — Divided (50+10) by 10 instead of (50−10)/10.
3 — Off by one; A=$140, B=$130. Not equal.
5 — A=$200, B=$210. Not equal.
4 — CORRECT.

🪤 **TRAP ALERT:** Dividing the SUM of join fees (60) by rate difference (10) instead of the DIFFERENCE of join fees (40).

💡 **KEY INSIGHT:** Break-even = (higher fixed cost − lower fixed cost) ÷ (rate of cheaper fixed − rate of more expensive fixed).

⚡ **FAST METHOD:** (50−10)/(40−30) = 40/10 = 4 months.

Q21. Correct Answer: D) 3

STEP-BY-STEP SOLUTION:

Parallel lines have the SAME slope.
First line slope = 3.
So k = 3.
y-intercepts differ (7≠−5), confirming parallel, not identical.

WHY EACH OPTION IS WRONG/RIGHT:

A) −3 — Product of slopes = 3×(−3)=−9≠−1. Not even perpendicular.
B) −1/3 — This is the PERPENDICULAR slope to 3.
C) 1/3 — Neither parallel nor perpendicular.
D) 3 — CORRECT: same slope = parallel.

🪤 **TRAP ALERT:** Confusing parallel (same slope) with perpendicular (negative reciprocal). Students pick B for perpendicular.

💡 **KEY INSIGHT:** Parallel = same slope. Perpendicular = slopes multiply to −1.

⚡ **FAST METHOD:** Read slope from y=3x+7 → slope=3. Match: k=3.

Q22. Correct Answer: C) 4

STEP-BY-STEP SOLUTION:

Parking total: 4 + 2h

Garage total: $6 + 1.5h$
Set equal: $4+2h = 6+1.5h$
$0.5h = 2 \rightarrow h = 4$
Verify: Parking=4+8=\$12. Garage=6+6=\$12 ✓

WHY EACH OPTION IS WRONG/RIGHT:

A) 2 — \$8 vs \$9.
B) 3 — \$10 vs \$10.50.
C) 4 — CORRECT.
D) 5 — \$14 vs \$13.50.

🪤 **TRAP ALERT:** Using total hours instead of ADDITIONAL hours (h). The first-hour charge is the flat fee.

💡 **KEY INSIGHT:** h = additional hours AFTER the first. Read the variable definition carefully.

⚡ **FAST METHOD:** $0.5h=2 \rightarrow h=4$.

Q23. Correct Answer: C) 3

STEP-BY-STEP SOLUTION:

Substitute x=0 into BOTH equations:
Eq1: $(k-2)(0)+3y=9 \rightarrow 3y=9 \rightarrow y=3$
Eq2: $2(0)+(k+1)y=12 \rightarrow (k+1)(3)=12 \rightarrow k+1=4 \rightarrow k=3$
Verify uniqueness with k=3: eq1 becomes $x+3y=9$, eq2 becomes $2x+4y=12 \rightarrow x+2y=6$.
Slopes: $-1/3 \neq -1/2$. Different slopes = exactly one solution ✓
Check: x=0, y=3 in eq2: 0+2(3)=6=6 ✓. In eq1 (with k=3): 0+9=9 ✓

WHY EACH OPTION IS WRONG/RIGHT:

A) 1 — $(1+1)(3)=6\neq12$.
B) 2 — $(2+1)(3)=9\neq12$.
C) 3 — CORRECT.
D) 5 — $(5+1)(3)=18\neq12$.

🪤 **TRAP ALERT:** Only substituting x=0 into ONE equation. You must use both to find k.

💡 **KEY INSIGHT:** x=0 solution → substitute into BOTH equations. First gives y, second gives k.

⚡ **FAST METHOD:** $3y=9 \rightarrow y=3$. Then $(k+1)(3)=12 \rightarrow k=3$.

Q24. Correct Answer: 20

STEP-BY-STEP SOLUTION:

Let x=liters of X (20%), y=liters of Y (50%).
Volume: $x + y = 60$
Acid: $0.20x + 0.50y = 0.30(60) = 18$
From volume: $x=60-y$. Substitute:
$0.20(60-y)+0.50y = 18$
$12-0.20y+0.50y = 18$
$0.30y = 6 \rightarrow y = 20$

Verify: x=40. Acid: 0.20(40)+0.50(20)=8+10=18=30% of 60 ✓

WHY EACH OPTION IS WRONG/RIGHT:

40 — Solved for x (Solution X) instead of y (Solution Y).
18 — Used the acid amount as the volume.
30 — Took 30% of 60 as the answer.
20 — CORRECT.

🪤 **TRAP ALERT:** Solving for x and reporting it as the answer. The question asks for Solution Y.

💡 **KEY INSIGHT:** Mixture problems: Volume equation + Concentration equation. Always check which solution the question asks about.

⚡ **FAST METHOD:** 💻 DESMOS: x+y=60 and 0.2x+0.5y=18. Find y at intersection.

Q25. Correct Answer: C) 5

STEP-BY-STEP SOLUTION:

Check coefficient ratio: 6/2 = 3 and −9/−3 = 3. Equal ✓
For no solution: coefficient ratios equal BUT constant ratio ≠ 3.
15/c must NOT equal 3, so c ≠ 5.
If c=5: 15/5=3 = coefficient ratio → infinitely many solutions (not no solution).
So c=5 CANNOT produce no solution.
All other values (3, 4, 7) produce no solution since 15/c ≠ 3.

WHY EACH OPTION IS WRONG/RIGHT:

A) 3 — 15/3=5≠3. No solution is possible ✓.
B) 4 — 15/4=3.75≠3. No solution is possible ✓.
C) 5 — CORRECT: 15/5=3 → all ratios equal → infinitely many, not no solution.
D) 7 — 15/7≈2.14≠3. No solution is possible ✓.

🪤 **TRAP ALERT:** Thinking c=5 gives no solution because 5 is "different." Actually c=5 makes ALL ratios equal → infinitely many.

💡 **KEY INSIGHT:** No solution: coeff ratios equal, constant ratio DIFFERENT. If constant ratio also equals → infinitely many instead.

⚡ **FAST METHOD:** Coeff ratio=3. If 15/c=3 → c=5 → infinitely many. So c=5 is impossible for no solution.

Q26. Correct Answer: A) −6

STEP-BY-STEP SOLUTION:

For no solution: coefficient ratios equal, constant ratio different.
x-coeff ratio: 5/10 = 1/2
y-coeff ratio must also = 1/2: −3/c = 1/2 → c = −6
Verify constant ratio: 7/20 = 0.35 ≠ 0.5 ✓
Different constant ratio confirms no solution.

WHY EACH OPTION IS WRONG/RIGHT:

A) −6 — CORRECT.
B) −3 — Ratio: −3/−3=1 ≠ 1/2. Different slopes → one solution.

C) 3 — Ratio: $-3/3=-1 \neq 1/2$. One solution.
D) 6 — Ratio: $-3/6=-1/2 \neq 1/2$. One solution.

🪤 **TRAP ALERT:** Getting c=6 by forgetting the negative. $-3/c=1/2$ means c must be NEGATIVE.

💡 **KEY INSIGHT:** Watch the sign: $-3/c = 1/2 \rightarrow c = -3\times2 = -6$. The negative in −3 carries through.

⚡ **FAST METHOD:** Set $-3/c = 5/10 = 1/2 \rightarrow c=-6$.

Q27. Correct Answer: A) 6

STEP-BY-STEP SOLUTION:

For infinitely many: eq2 = multiplier × eq1.
y-coefficient ratio: $6/3 = 2$ (multiplier = 2).
x-coefficient: $2(a+1) = 4 \rightarrow a+1=2 \rightarrow a=1$
Constant: $2b = 12 \rightarrow b=6$
$ab = 1\times6 = 6$

WHY EACH OPTION IS WRONG/RIGHT:

A) 6 — CORRECT.
B) 12 — Set b=12 without dividing by 2.
C) 18 — Used a=3 instead of a=1.
D) 24 — Used a=4 (the x-coeff from eq2) without solving.

🪤 **TRAP ALERT:** Using 4 as the value of a (it's the coefficient in eq2, not the answer). Solve $2(a+1)=4$.

💡 **KEY INSIGHT:** Multiplier = 6/3 = 2. Apply to all: $2(a+1)=4 \rightarrow a=1$, $2b=12 \rightarrow b=6$.

⚡ **FAST METHOD:** Multiplier=2. a=1, b=6. ab=6.

Q28. Correct Answer: 35

STEP-BY-STEP SOLUTION:

Night 1: $120f + 80b = 8{,}800 \rightarrow$ divide by 40: $3f + 2b = 220$ ...(A)
Night 2: $90f + 150b = 9{,}750 \rightarrow$ divide by 30: $3f + 5b = 325$ ...(B)
Subtract (A) from (B): $3b = 105 \rightarrow b = 35$
$f = (220-70)/3 = 150/3 = 50$
Verify night 1: $120(50)+80(35) = 6{,}000+2{,}800 = 8{,}800$ ✓
Verify night 2: $90(50)+150(35) = 4{,}500+5{,}250 = 9{,}750$ ✓

WHY EACH OPTION IS WRONG/RIGHT:

$50 — This is the floor seat price, not balcony.
$46 — Averaging total revenue by total seats (wrong method).
$30 — Arithmetic error in division.
$35 — CORRECT.

🪤 **TRAP ALERT:** Dividing total revenue by total seats to get an average price. You need the system of equations to find each individual price.

💡 **KEY INSIGHT:** Two unknowns require two equations. Set up both nights as separate equations.

⚡ **FAST METHOD:** 💻 DESMOS: Enter 120f+80b=8800 and 90f+150b=9750. Find b at intersection.

Q29. Correct Answer: C) 2

STEP-BY-STEP SOLUTION:

Set equal: $2x^2-8x+6 = 2x-2$
$2x^2-10x+8 = 0$
$x^2-5x+4 = 0$
$(x-1)(x-4) = 0$
$x=1$ or $x=4$ → TWO solutions
Verify: $x=1$→$y=0$ in both. $x=4$→$y=6$ in both ✓
Discriminant check: $b^2-4ac = 25-16 = 9 > 0$ → 2 real solutions ✓

WHY EACH OPTION IS WRONG/RIGHT:

A) 0 — Discriminant is positive (9>0), so solutions exist.
B) 1 — Would require discriminant=0, but it's 9.
C) 2 — CORRECT.
D) Infinitely many — Only if the equations were identical, which they aren't.

🪤 **TRAP ALERT:** Making an arithmetic error in the quadratic that falsely gives a negative discriminant.

💡 **KEY INSIGHT:** Line-parabola system: subtract → quadratic. Discriminant > 0 → 2 solutions; = 0 → 1; < 0 → 0.

⚡ **FAST METHOD:** Discriminant: $(-5)^2-4(1)(4)=25-16=9>0$ → 2 solutions. No need to factor.

Q30. Correct Answer: B) 2025

STEP-BY-STEP SOLUTION:

Set populations equal:
$4{,}000 + 300t = 6{,}500 - 200t$
$500t = 2{,}500$
$t = 5$ years after 2020
Year = 2020 + 5 = 2025
Verify: A=4,000+1,500=5,500. B=6,500−1,000=5,500 ✓

WHY EACH OPTION IS WRONG/RIGHT:

A) 2024 — t=4: A=5,200, B=5,700. Not equal.
B) 2025 — CORRECT.
C) 2026 — t=6: A=5,800, B=5,300. A surpassed B.
D) 2027 — Too late.

🪤 **TRAP ALERT:** Writing t=5 as the answer instead of adding it to the base year 2020.

💡 **KEY INSIGHT:** "t years since 2020" → calendar year = 2020 + t.

⚡ **FAST METHOD:** 500t=2500→t=5→year=2025.

TOPIC SUMMARY — Systems of Linear Equations

Common Mistakes:

1. Solving for x when the question asks for y (or vice versa), or forgetting to compute the final expression (like 2a or x+y).

2. Sign errors when subtracting equations — every term changes sign.

3. Confusing no solution (parallel) with infinitely many (identical). No solution: $a_1/a_2 = b_1/b_2 \neq c_1/c_2$. Infinitely many: ALL ratios equal.

Must Remember:

1. Opposite coefficients → ADD. Same coefficients → SUBTRACT.

2. Infinitely many: ratio of ALL three must match. No solution: coefficient ratios match, constant ratio differs.

3. Word problems: write the count equation first, then the money/mixture equation.

Expected on DSAT: 3–5 questions per test. One no-solution/infinitely-many, one word problem, one find-the-value.

🖥️ DESMOS STRATEGY: Type both equations into Desmos and click the intersection point. Works for any system in under 20 seconds. Practice this — it is the single biggest time-saver on the DSAT math section.